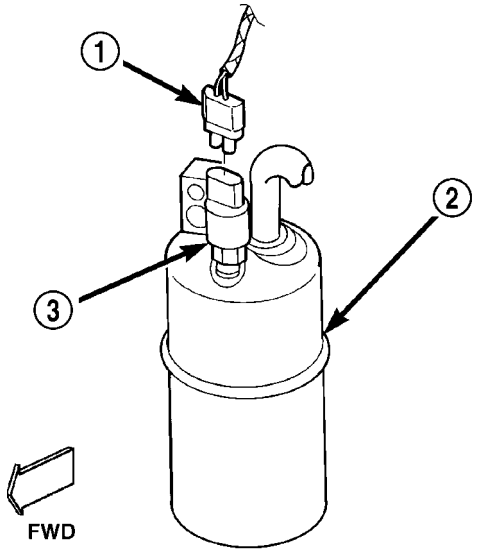

Fig. 19 A/C Low Pressure Switch - Typical

1 - WIRE HARNESS CONNECTOR
2 - ACCUMULATOR
3 - A/C LOW PRESSURE SWITCH

INSTALLATION

NOTE: Use only the specified O-ring as it is made of special material for R-134a. Use only refrigerant oil of the type required for the A/C compressor.

(1) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the A/C low pressure switch fitting.

(2) Install the A/C low pressure switch on the accumulator. Hand-tightened the switch securely.

(3) Connect the wire harness connector to the A/C low pressure switch.

(4) Reconnect the negative battery cable.

A / C PRESSURE TRANSDUCER

DESCRIPTION

The A/C pressure transducer (Fig. 20) is a switch that is installed on a fitting located on the A/C discharge line when equipped with the 2.4L gasoline engine. An internally threaded fitting on the A/C pressure transducer connect it to the externally threaded Schrader-type fitting on the A/C discharge line. A rubber O-ring seals the connection between the A/C pressure transducer and the discharge line fitting. The A/C pressure transducer is connected to the vehicle electrical system by a molded plastic connector with three terminals.

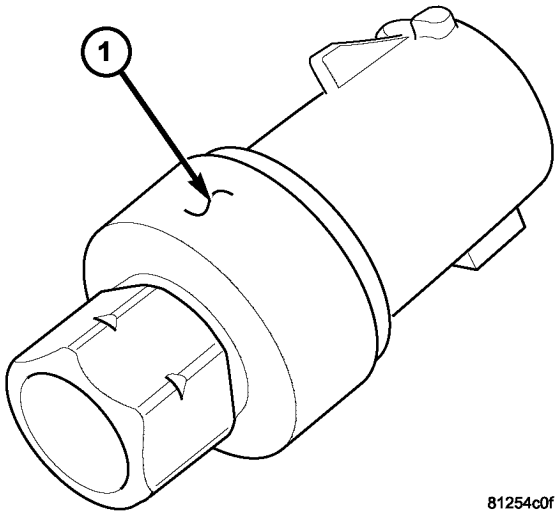

Fig. 20 A/C Pressure Transducer

1 - A/C PRESSURE TRANSDUCER

OPERATION

The A/C pressure transducer monitors the pressures in the high side of the refrigerant system through its connection to a fitting on the A/C discharge line. The A/C pressure transducer will change its internal resistance in response to the pressures it monitors. A Schrader-type valve in the discharge line fitting permits the A/C pressure transducer to be removed or installed without disturbing the refrigerant in the A/C system.

The powertrain control module (PCM) provides a five volt reference signal and a sensor ground to the A/C pressure transducer, then monitors the output voltage of the A/C pressure transducer on a sensor return circuit to determine refrigerant pressure. The PCM is programmed to respond to the A/C pressure transducer and other sensor inputs by controlling the operation of the A/C compressor clutch and the radiator cooling fan to help optimize A/C system performance and to protect the system components from damage PCM will disengage the A/C compressor clutch when high side pressure rises above 2971 kPa (431 psi) or fall below 206 kPa (30 psi). The A/C pressure transducer input to the PCM will also prevent the A/C compressor clutch from engaging when ambient temperatures are below about 1.0° C (50° F) due to the pressure/temperature relationship of the refrigerant.

The A/C pressure transducer is diagnosed using a DRBIII® scan tool. Refer to the appropriate diagnostic information.

The A/C pressure transducer cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.